Technology Stack

Date	16 October 2026
Team ID	
Project Name	Personalized Travel Itinerary Planner
Maximum Marks	2 Marks

Define Your Technology Stack:
Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side** *(e.g., React, Angular, HTML/CSS)*	*React 18 + Vite (JavaScript, CSS3)*	• *Fast Single-Page Application (SPA) rendering with rapid Vite HMR.* • *Reusable modular components for vibe sliders, cards, and daily timelines.* • *Responsive, mobile-first glassmorphism design.*

2	**Backend / Server-Side** *(e.g., Node.js, Python/Django, Java)*	*Python 3.11 + FastAPI (Uvicorn ASGI)*	*• High-throughput asynchronous REST API endpoints.* *• Automatic Pydantic request validation and interactive Swagger docs.* *• Seamless integration with Python AI SDKs and route optimization scripts.*
3	**Database / Data Storage** *(e.g., MySQL, MongoDB, PostgreSQL)*	*Supabase (PostgreSQL 15)*	*• Robust relational data integrity for trips, days, and activities.* *• Row-Level Security (RLS) ensures users only access their own itineraries.* *• Serverless cloud PostgreSQL with fast JSON query support.*
4	**Cloud / Hosting / Deployment** *(e.g., AWS, Azure, Heroku, Docker)*	*Vercel (Frontend) & Render (Backend)*	*• Vercel: Global Edge CDN delivering ultra-fast frontend loads and automatic preview builds.* *• Render: Scalable cloud hosting for the Python FastAPI server with automatic Git deployments.*
5	**Version Control & CI/CD**	*Git & GitHub*	*• Distributed version control for collaborative code tracking and branching.* *• Automated Git webhook triggers enabling instant CI/CD deployment to Vercel and Render on git push.*

	(e.g., GitHub, GitLab, Jenkins)		
6	**Third-Party APIs / Other Tools** *(e.g., Stripe, Twilio, Google Maps)*	*Google Gemini 2.5 Flash API, 2-Opt TSP Algorithm, iCalendar (.ics)*	• *Google Gemini: Generates natural-language personalized schedules from traveler vibes.* • *2-Opt TSP & Haversine: Reorders stops to minimize transit travel time.* • *iCalendar (.ics): 1-click export to Google/Apple Calendar.*